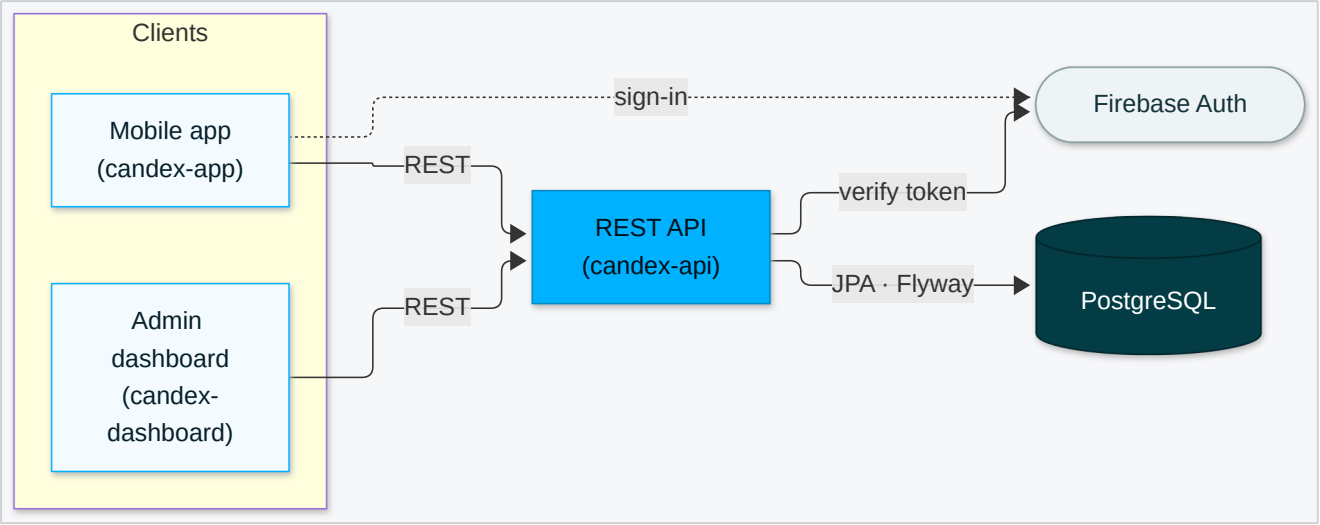
# Working on Candex

This page orients you if you are going to work on Candex. For the product itself — what it is and how it feels to use — start with Understanding Candex.

## The system at a glance

Candex is not one app but a few pieces working together. Users touch the **mobile app**. An **admin dashboard** keeps the drink catalogue accurate. Both talk to a single **REST API**, which holds the domain and stores everything in **PostgreSQL**. Sign-in is handled by **Firebase**.




## Repositories

| Repository | Role | Status |
| --- | --- | --- |
| `candex-app` | Mobile client (Angular + Ionic + Capacitor) | Active |
| `candex-dashboard` | Admin dashboard for managing the catalogue | Active |
| `candex-api` | REST API and domain model, backed by PostgreSQL | Active |
| `candex-handbook` | This documentation site (Astro Starlight) | Active |
| `candex-crawler` | Crawler that gathers drink and catalogue data | Active |
| `candex-ai` | AI module (Python) | Early / placeholder |
| `candex-backups` | Firestore backup tooling | Deprecated — predates the move to PostgreSQL |
| `candex-api-legacy` | Previous API (Node.js / Express) | Deprecated |

> **Note**
>
> `candex-crawler` lives in its own repository (outside the main workspace); its role is summarised here from its purpose and will be detailed in its own handbook.

# Tech stack

**Mobile app** — `candex-app`



- **TypeScript**
- **NgRx** — Store • Effects • Entity
- **RxJS**
- **Firebase** — Auth

**Admin dashboard** — `candex-dashboard`

- **Angular** — v20
- **Ionic** — v8
- **TypeScript**
- **Angular Material**
- **Firebase** — Auth

**Back-end** — `candex-api`

- **Spring Boot** — v3.5
- **Java** — 21
- **Maven** — build (wrapper)
- **PostgreSQL**
- **Flyway** — migrations
- **MapStruct** — mapping
- **Spring Security** — OAuth2
- **OpenAPI** — springdoc

**Quality & delivery**

- **Docker** — Compose
- **JUnit** — + Mockito
- **Testcontainers**
- **JaCoCo** — coverage
- **ESLint**
- **Prettier**
- **Cypress** — e2e

# Where things live

- **Domain & API** — `candex-api`. Entities and business rules in `src/main/java`; schema in PostgreSQL, evolved through Flyway migrations under `src/main/resources/db/migration`.
- **Mobile UI** — `candex-app` (Angular + Ionic, NgRx for state).
- **Admin UI** — `candex-dashboard` (Angular + Ionic + Angular Material).

The codebase is the source of truth: when documentation and code disagree, the code wins and the docs are corrected.

# Running it locally

Candex runs as three apps — the mobile app and the dashboard (Ionic/Angular, run with npm) and the API (Maven + Spring Boot, with PostgreSQL), each in three environments (local, dev, prod). The full setup — prerequisites, cloning, environment files, the database, and the per-app commands — lives in the **Local Installation** handbook *(planned)*.

# Where to go next

- Set up your machine → Local Installation *(planned)*
- Understand how the pieces fit, and why → Architecture & Overview *(planned)*
- The precise domain language → Glossary / Domain *(planned)*

**Your first step:** clone the repositories, then follow Local Installation to get the API and at least one front-end running against a local database.